Benchmark results

NewTOS on Raspberry PI 4

unknown AArch64 TOS 4.04 1920x1080 256c blit none

115 of 115 tests run, 1.0 s each. Values are operations per second (higher is better). Bars compare tests within each group.

Groups

VDI primitives	38 tests run
AES windows & events	21 tests run
Alignment (blitter)	20 tests run
Full-screen windows	17 tests run
GDOS fonts	14 tests run

VDI primitives

Test	ops/s	Note
Line, 2 points	22545	
Polyline, 32 points	769.4	
Wide line (7px, round)	4014.7	
Dashed line	20572	
User line style	21870	
XOR line	11376	
Markers x16	22967	
Filled rect 32x32	93621	
Filled rect, full window	578.2	
Bar, hatched + border	51199	
Pattern fill rect	92738	
User pattern fill	93621	
XOR filled rect	121063	
Polygon, 8 points	17553	
Filled circle	2618.6	
Filled ellipse	2996.0	
Arc	21152	
Pie slice	2655.6	
Elliptical arc	22194	
Elliptical pie	3025.6	
Rounded box	40353	
Filled rounded box	9659.4	
Contour fill 64x40	7383.6	
Text, 15 chars	17225	
Text, 84 chars	2870.0	
Text with effects	7455.3	
Text rotated 90	15359	
Justified text	10137	
Text extent query	396263	
Blit scr->scr 64x64	22414	
Blit scroll window	119.2	
Blit mem->scr 64x64	27835	
Blit scr->mem 64x64	24650	
Mono blit (vrt_cpyfm)	13460	
Transform form 64x64	n/a	not supported here
Get pixel	504122	
Hide/show mouse	20176	
Attribute calls x3	170827	

Open/close workstation	46810	

AES windows & events

Test	ops/s	Note
Window create+delete	151702	
Window open+close	220.7	
Window move	368.2	
Window resize	844.3	
Window top (2 windows)	447.0	
Set window title	1379.4	
Set info line	1393.0	
Set slider pos+size	2026.7	
Scroll window content	118.4	
wind_get work area	329175	
Rectangle list walk	147455	
wind_calc	300622	
wind_update pair	169223	
objc_draw dialog	3727.1	
objc_find	286321	
form_dial redraw 64x48	5878.4	
Menu bar redraw	4226.6	
graf_mkstate	330685	
graf_mouse off+on	19319	
Event poll (evnt_multi)	141783	
Message round trip	60234	

Alignment (blitter)

Test	ops/s	Note
Fill 64x64, aligned	114116	
Fill 64x64, x+3	46810	
Blit 64x64, aligned	24211	
Blit 64x64, src+3 dst+11	21251	
Mem->scr 64x64, aligned	27971	
Mem->scr 64x64, x+3	27701	
Mono blit, aligned	13114	
Mono blit, x+3	13523	
Text, x on 8 px	18069	
Text, x+3	16831	
Scroll up, aligned edges	123.9	
Scroll up, edges x+5	123.9	
Scroll left 16 px	124.5	
Scroll left 3 px	123.9	
Move window, aligned	354.6	work x mod 16 = 0
Move window, off-grid	351.3	work x mod 16 = 5
Open/close win, aligned	227.6	work x mod 16 = 0
Open/close win, off-grid	226.5	work x mod 16 = 5
Scroll window, aligned	400.9	work x mod 16 = 0
Scroll window, off-grid	395.2	work x mod 16 = 5

Full-screen windows

Test	ops/s	Note
Open/close full window	135.5	work 1896x1018
Full <-> normal size	242.8	
Top full / test window	292.6	work 1896x1018
Fill full window	377.3	work 1896x1018
Scroll up 1 line	81.6	work 1896x1018
Scroll left 16 px	80.5	work 1896x1018
Scroll left 8 px	79.8	work 1896x1018
Scroll left 3 px	79.8	work 1896x1018
Scroll text 8pt	81.6	system font
Scroll text 10pt	81.6	system font
Scroll text 12pt	81.6	got 10pt, system font
Scroll text 14pt	81.6	got 10pt, system font
Scroll text 18pt	80.5	system font
Scroll text 24pt	79.8	got 20pt, system font
Scroll text 36pt	79.8	got 20pt, system font
Scroll text 48pt	79.8	got 20pt, system font
Scroll, size per line	80.9	system font

GDOS fonts

Test	ops/s	Note
Text 6pt	35963	got 8pt, system font
Text 8pt	36140	system font
Text 10pt	12557	system font
Text 12pt	12498	got 10pt, system font
Text 14pt	12557	got 10pt, system font
Text 18pt	6729.5	system font
Text 24pt	3461.8	got 20pt, system font
Text 36pt	3478.6	got 20pt, system font
Text 48pt	3495.6	got 20pt, system font
Text 72pt	3445.1	got 20pt, system font
Text 10pt bold	12382	system font
vst_point, 8 sizes	391258	
Font switch + vst_point	n/a	not supported here
Arbitrary size + text	n/a	not supported here
Scroll, mixed sizes	115.5	system font
Text extent at 12pt	399984	
Enumerate font names	n/a	not supported here
Load+unload fonts (x3)	n/a	not supported here

Configuration

Machine

```
Description     : NewTOS on Raspberry PI 4
Machine type    : unknown
CPU             : AArch64 (native)
FPU             : none reported
Blitter         : not present
ST RAM          : 14336 KB total
TT/Fast RAM     : 729088 KB total
```

Operating system

```
TOS             : 4.04  (11/07/2026 m/d/y)
Country / video : USA, NTSC
GEMDOS          : 0.48
MiNT            : not running
Largest free    : ST 10240 KB
                  TT/Fast 715424 KB
```

AES / environment

```
AES version     : 3.40
Multitasking    : no (max apps 1)
Theme pens      : green 166, red 122, blue 149, cyan 155, highlight 106
Application id  : 0
appl_getinfo    : not available
Desktop area    : 0,19 1920x1061
AES char / box  : 8x16 / 19x19
Current dir     : C:\APPS\FBENCH
PATH            : /dev/c/
```

Display

```
Resolution      : 1920x1080, Getrez 4 (TT Medium)
Colours         : 256 on screen, 16 planes
Palette         : 0 (direct colour)
Lookup table    : no (true colour)
Pixel size      : 278 x 278 microns
Screen memory   : 4147200 bytes at $3dbf5000
VDI handles     : physical 1, ui 3 (bench ws opened only for runs)
VDI max vertices: 1024
Text rotation   : 90 degree steps
Writing modes   : 4
```

GDOS / fonts

```
GDOS            : none (system font only)
Fonts           : 1 (0 loaded by GDOS)
  > id     1  6x6 system font
```

Cookie jar

```
NTOS _MCH _IDT _AKP EdDI
```